Marathwada Shikshan Prasarak Mandal's

Deogiri Institute of Engineering and Management Studies, Chhatrapati Sambhajinagar

Project Report on

Scholarship Navigator – Smart Student Funding Assistant

Submitted By

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Dr. Babasaheb Ambedkar Technological University Lonere (M.S.)



Department of Computer Science and Engineering

Deogiri Institute of Engineering and Management Studies, Chhatrapati Sambhajinagar

(2026- 2027)

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Akanksha Popat Jadhav (CS4101)
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In partial fulfillment of

Bachelor of Technology

(Computer Science and Engineering)

Guided By

Mrs .Sugandha Nandedkar

Department of Computer Science and Engineering

Deogiri Institute of Engineering and Management Studies,

Chhatrapati Sambhajinagar

(2026- 2027)

CERTIFICATE

This is to certify that, the Project entitled "**Scholarship Navigator – Smart Student Funding Assistant**" submitted by **Akanksha Popat Jadhav (CS4101), Vaishnavi Rajendra Thombre (CS4102), Vaishnavi Sanjay Untwal (CS4143)** is a bonafide work completed under my supervision and guidance in partial fulfillment for award of Bachelor of Technology (Computer Science and Engineering) Degree of Dr. Babasaheb Ambedkar Technological University, Lonere.

Date:
Place: Chhatrapati Sambhajinagar

Project Guide	Project Co-ordinator
Mrs. Sugandha Nandedkar	**Ms. R. R. Galgali**
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Head of Department	Director

DECLARATION

This is to certify that, the Project Report entitled (TNRB 12), "**Scholarship Navigator – Smart Student Funding Assistant**" Submitted by **Akanksha Popat Jadhav (CS4101), Vaishnavi Rajendra Thombre (CS4102), Vaishnavi Sanjay Untwal (CS4143)** is a bonafide work completed under my supervision and guidance in partial fulfillment for award of bachelor's degree in Computer Science and Engineering of Deogiri Institute of Engineering and Management Studies, Chhatrapati Sambhajinagar under Dr. Babasaheb Ambedkar Technological University, Lonere.

Place: Chhatrapati Sambhajinagar
Date:

Mrs Sugandha Nandedkar
Guide

External Examiner

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CHAPTER 1: INTRODUCTION

1.1 Background

Education is an important foundation for personal development, employment, innovation, and social progress. Higher education can help students develop specialized knowledge and professional skills, but the cost of tuition, accommodation, travel, books, equipment, and other academic requirements can create a substantial financial burden. This burden may be especially difficult for students from economically disadvantaged households or for students whose families have limited access to reliable information about financial assistance.

Scholarships are one of the ways in which governments, educational institutions, charitable trusts, foundations, companies, and other organizations support students. Depending on the scheme, assistance may be based on academic merit, family income, social category, disability, gender, location, field of study, course level, or a combination of these factors. Some schemes provide tuition support, while others provide a fixed amount, maintenance assistance, or support for particular educational expenses. Because the rules and application periods differ, students need to understand each scheme before deciding whether to apply.

A large amount of scholarship information is now published online. Official government portals, university websites, private foundations, corporate social responsibility programmes, and education-related websites may each publish different opportunities. This online availability is useful, but it can also make the discovery process difficult. Information is spread across different websites, presented in different formats, and updated at different times. A student may have to visit many pages, compare eligibility rules, check the required documents, identify the application deadline, and locate the official application link.

The National Scholarship Portal (NSP) in India is an important example of a centralized public platform. The National Informatics Centre describes it as a one-stop solution supporting scholarship application, processing, sanction, and disbursal, with features intended to simplify access and improve transparency. However, students may also need to explore state-specific, institution-specific, charitable, and private opportunities that are published through other channels. A student who relies on only one website may therefore overlook other relevant opportunities.

ScholarNexus is proposed as a scholarship navigator that helps students discover potentially suitable scholarships through a centralized interface. Its purpose is to organize information from available sources, compare scholarship conditions against a student's profile, and present relevant opportunities in a more understandable form. The system is intended to reduce repetitive manual searching and make important details—such as minimum marks, annual family income limits, eligible courses, scholarship benefits, required documents, and deadlines—easier to identify.

1.2 Problem Statement

Students seeking financial assistance for higher education face difficulty finding suitable scholarships because information is scattered across multiple websites and portals. They must manually search different platforms and check varying eligibility criteria, which is time-consuming and inefficient. As a result, students may miss suitable scholarship opportunities and important application deadlines. Therefore, there is a need for a centralized and personalized system that simplifies scholarship discovery, eligibility checking, and recommendation.

1.3 Need for the System

The main need for ScholarNexus arises from the gap between the availability of scholarship information and a student's ability to find, interpret, and act on that information. A scholarship may exist online but still be difficult to discover if it is published on an unfamiliar website, described using complex wording, or listed among many unrelated opportunities. The student must also determine whether the scheme applies to their course, academic year, location, income group, and other circumstances.

A centralized navigator can provide a single point from which students search and compare opportunities. Personalization can further narrow the results by using relevant profile attributes. For example, if a scholarship requires a particular course level, minimum percentage, state of residence, and income ceiling, the system can compare these requirements with the student's entered details and indicate whether the student appears to meet them. Such a result should be presented as preliminary guidance rather than a guarantee of selection, because the final decision belongs to the scholarship provider and may depend on document verification or additional rules.

The system also needs to present information in a structured format. Scholarship names, providers, benefits, minimum marks, income limits, eligible courses, deadlines, document requirements,

source URLs, and last-checked information should be represented consistently wherever available. A clear record helps students compare options without repeatedly opening many pages. If a detail is not published or could not be verified, it should be marked as unavailable or requiring confirmation instead of being guessed.

1.4 Aim and Objectives

The aim of ScholarNexus is to develop a centralized scholarship discovery and recommendation system that helps students identify financial-aid opportunities relevant to their educational and financial profiles.

The objectives are:

- To provide a unified interface for exploring scholarship opportunities from multiple available sources.
- To collect and organize scholarship details in a structured format.
- To use a crawler or data-collection component to discover relevant pages where permitted and technically feasible.
- To extract scholarship information from source pages and normalize it for consistent display.
- To allow students to enter profile details relevant to scholarship eligibility, such as course, academic level, marks, state, and family income, where required.
- To compare student profile information with the stated eligibility conditions of scholarships.
- To recommend or filter scholarships that appear relevant to each student.
- To display important information, including scholarship benefits, deadlines, eligibility conditions, required documents, and official application links when available.
- To reduce manual effort and help students prioritize opportunities that deserve further verification.
- To encourage responsible use of public information and make the source of each scholarship listing visible.

1.5 Scope of the Project

The proposed system focuses on scholarship discovery, information organization, preliminary eligibility matching, and recommendation. It is not intended to replace the official application process or the scholarship provider's verification process. Students should be able to browse listings, review key conditions, and follow the relevant official link for the authoritative details and application procedure.

The scope may include a student-facing interface, a scholarship information store, a data collection or crawler module, an extraction and normalization process, a matching mechanism, and administrative facilities for reviewing records. Depending on the implemented version, the application may also provide search, filtering, a chatbot or help interface, saved scholarships, deadline reminders, and an administrative dashboard. Only features that are implemented and tested should be described as completed in the final project demonstration.

The system is especially relevant to students pursuing undergraduate, postgraduate, diploma, and other eligible educational programmes. It can be extended to support more institutions, disciplines, regions, languages, and scholarship providers. The quality of recommendations will depend on the completeness, accuracy, and freshness of the stored scholarship data and on the correctness of the eligibility rules.

1.6 Proposed System Overview

At a high level, ScholarNexus follows a data-to-recommendation workflow. First, the system identifies relevant scholarship sources. A crawler may visit approved pages and collect links or page content. An extraction component then identifies fields such as scheme title, provider, eligibility criteria, award amount, deadline, and application URL. The extracted details are cleaned and converted into a consistent structure before being stored. The student enters profile details, and the matching component compares the profile against the available scholarship rules. Finally, the interface displays matching or potentially matching opportunities, including the evidence needed for the student to review each result.

The workflow can be summarized as: source discovery → page collection → data extraction → data validation and normalization → database storage → student profile input → eligibility matching → ranked or filtered results → official source verification and application. The stages

should be designed so that a failure in one source does not corrupt unrelated records. Records with missing or conflicting information should be flagged for review.

A crawler and a scraper have related but different functions. A crawler discovers and visits pages by following links or a defined source list. A scraper extracts selected information from those pages. For ScholarNexus, a crawler can locate relevant scholarship pages, while an extraction process can read the scholarship fields from a page. If the system only reads a fixed list of URLs, it is more accurate to describe that part as scheduled page fetching or scraping rather than claiming broad web crawling. The final report should reflect the actual implementation.

1.7 Significance of the Project

For students, ScholarNexus can reduce the time required to find relevant scholarships and make eligibility conditions easier to compare. A structured result may help a student notice a minimum percentage requirement, an income ceiling, or a deadline before beginning an application. It can also support students who are unfamiliar with scholarship terminology or who do not know which organizations publish opportunities.

For educational institutions and student-support teams, a centralized listing can provide a starting point for guiding students toward relevant funding opportunities. For developers and researchers, the project demonstrates the application of web technologies, data extraction, database design, rule-based matching, and user-centered information retrieval to a practical educational problem.

The project also has a broader inclusion objective. When information is easier to locate and understand, students may be better positioned to explore financial support that could otherwise remain unnoticed. Nevertheless, technology alone cannot guarantee access or award outcomes. The system should communicate limitations clearly, protect profile information, and direct students to authoritative sources.

CHAPTER 2: LITERATURE SURVEY

2.1 Introduction to the Literature Survey

A literature survey examines existing platforms, research studies, methods, and technical approaches related to a project. It helps establish what has already been developed, what methods are commonly used, and which limitations remain relevant. For ScholarNexus, the survey covers centralized scholarship portals, rule-based eligibility checking, recommender systems, web crawling and scraping, information quality, and personalized support for students.

The literature indicates that scholarship discovery is connected to several established areas of computer science. Centralized portals address the organization of services and records. Recommender systems filter a large collection of items according to user needs. Rule-based systems compare facts against explicit conditions. Web crawlers locate pages, while scrapers extract structured data from web content. Combining these capabilities can support a practical scholarship navigator, provided that data accuracy, source transparency, privacy, and maintainability are treated as core requirements.

2.2 Centralized Scholarship Portals

The National Scholarship Portal (NSP) is a significant Indian example of centralized scholarship administration. The National Informatics Centre describes the NSP as a one-stop solution for processes ranging from application receipt and processing to sanction and disbursal. Its stated benefits include bringing scholarship information under one umbrella, providing an integrated application process, supporting eligibility-based scheme suggestions, and improving monitoring and transparency [1]. The official portal also provides scheme information and eligibility-related services for students [2].

These features demonstrate the value of centralized digital access. A common portal can reduce the need for students and administrators to use disconnected workflows for schemes that participate in the platform. It also illustrates the importance of structured scheme information, eligibility configuration, verification stages, and application-status information.

However, centralized administration and broad scholarship discovery are not identical goals. Some opportunities are published by individual colleges, charitable trusts, private foundations,

companies, or local organizations outside a particular portal. A student may therefore need to consult several sources even when a national portal is available. ScholarNexus is intended to serve as a discovery and comparison layer across available sources, while continuing to direct applicants to official provider pages for final verification and application.

2.3 Rule-Based Scholarship Eligibility Systems

A rule-based system represents domain knowledge as conditions that can be evaluated against input data. In scholarship matching, rules may include a minimum academic percentage, a maximum family income, a course or degree requirement, a residence condition, a category requirement, or a specific institution. If a scheme requires all conditions to be satisfied, the system can evaluate each condition and report which ones match, fail, or remain unknown.

An example of rule-based educational support appears in the 2014 study 'Rule-based Expert Systems for Supporting University Students'. The authors describe educational expert systems, including a system that suggests scholarships to undergraduate students based on eligibility, implemented using Oracle Policy Automation [3]. This work is relevant because it shows how scholarship rules can be encoded to assist student decision-making.

For ScholarNexus, rule-based matching is practical because many scholarship criteria are explicit and can be represented as structured fields. A transparent rule engine can explain why a scholarship appears in the results—for example, the student's course matches the scheme, the marks exceed the stated minimum, and the income value is within the published limit. Explainability is important because students should be able to review the conditions rather than receive an unexplained score.

The main limitation is that eligibility rules are not always written in a uniform way. One provider may state an annual family income limit, while another may describe a different income measure or require a particular certificate. Some schemes have exceptions, renewal rules, or conditions that cannot be reduced to a simple comparison. The system must therefore distinguish clear matches from incomplete or ambiguous records and should not infer missing rules.

2.4 Recommendation Systems

A recommender system is an information-filtering approach that selects or ranks items likely to be useful to a particular user. Recommenders are widely used to manage large collections of products,

media, documents, courses, and other resources. In a scholarship setting, the items are scholarship schemes and the user's profile provides context for determining relevance.

Zhang and colleagues published a broad literature survey of scholarly recommendation systems in 2023. Their review discusses recommendation tasks involving literature, collaborators, conferences, journals, datasets, and grant funding. The authors report that content-based filtering is commonly used in scholarly recommendation, while collaborative filtering is more prominent in some other recommendation tasks. They also identify relatively limited research in some funding-related recommendation areas and note that user-feedback analysis is uncommon in the reviewed literature [4].

A systematic review of course recommendation systems likewise identifies personalization as important in educational decision-making and discusses challenges such as a large, changing set of choices and the cold-start problem for new users [5]. A separate systematic review of recommender systems for higher-education choices reports that hybrid approaches are often effective, while also noting that many evaluated studies rely on laboratory settings rather than real-world deployment [6]. These findings are relevant to ScholarNexus because scholarship recommendations must work with changing information and students who may be using the system for the first time.

Several recommendation approaches can be considered. Content-based recommendation compares scholarship attributes with the student's profile. Rule-based filtering checks mandatory eligibility conditions. Collaborative filtering uses patterns from multiple users, such as preferences or interactions, but it generally requires enough user activity and careful privacy design. A hybrid approach can combine eligibility rules with relevance ranking, search terms, course interests, location, or scholarship type.

For an initial student project, eligibility filtering followed by transparent ranking is a reasonable design. Mandatory criteria should be evaluated first so that clearly ineligible schemes are not presented as strong matches. Remaining results can be ranked according to how closely the scholarship fits the student's preferences or profile. The interface should explain that ranking indicates relevance, not the probability of being awarded the scholarship.

2.5 Content-Based Filtering

Content-based filtering recommends items by comparing item attributes with a user profile or stated preferences. For a scholarship, attributes may include the eligible course, level of study, state, academic performance threshold, income limit, scholarship category, award amount, and provider. The student profile contains corresponding values where relevant. The system compares the two and identifies schemes that meet the mandatory conditions or share preferred characteristics.

Ferdiansyah and Suyatno's paper, 'Scholarship Recommendation System Using Content-Based Filtering Approach', describes a web-based scholarship recommendation system using content-based filtering and Euclidean distance, with precision testing as part of its evaluation [7]. This work directly relates to the proposed project because it applies recommendation concepts to the problem of finding scholarships that fit an individual educational profile.

Content-based methods are useful when the system has item descriptions and a student profile but little or no interaction history. This is particularly relevant to a new platform: it can offer recommendations to a first-time user without waiting for many students to rate or select scholarships. The approach can also be explained by referring to the attributes that influenced a match.

Its limitations include dependence on the quality of item metadata and the risk of over-filtering. If a scholarship record lacks the correct course level or income threshold, the comparison may be misleading. If the system only recommends items similar to those already selected, it may also fail to expose users to opportunities they did not know to search for. These limitations can be reduced through source validation, broad search and filter options, explicit unknown states, and periodic review of the stored data.

2.6 Web Crawling

Web crawling is the process of systematically discovering and fetching pages from the web or from a defined set of sources. A crawler commonly begins with known starting URLs, retrieves pages, identifies links, and decides which links should be visited next. Crawling is a foundational component of search engines and can be used to discover pages containing relevant information.

Najork's survey of web crawling describes the area as more complex than simply following links. It discusses challenges such as managing large data structures and deciding how frequently to

revisit pages whose content changes over time [8]. Bhatia and Rattan's survey of web crawlers for information retrieval reviews a range of crawling techniques and the role of crawlers in gathering pages for indexing and retrieval [9]. These studies highlight that crawler design must consider scale, page selection, freshness, and efficient retrieval.

In ScholarNexus, a crawler can be used to discover scholarship pages from a configured set of official or otherwise appropriate sources. It can prioritize known provider websites, follow relevant links, avoid duplicate URLs, and store source references for later extraction. A simple student implementation may use a controlled list of seed URLs rather than attempting to crawl the entire web. This makes the system more manageable and improves the ability to verify source quality.

A crawler alone does not understand whether a page contains a scholarship, nor does it automatically extract accurate eligibility details. It primarily helps locate and retrieve pages. The subsequent extraction, validation, and normalization stages are needed to turn page content into useful scholarship records.

2.7 Web Scraping and Data Extraction

Web scraping refers to extracting selected information from web pages and converting it into a structured representation. Depending on the source, extraction may use HTML elements, page metadata, tables, structured data, documents, or browser automation. A scraping module may retrieve a scholarship title, provider, eligibility text, amount, deadline, required documents, and application link.

A 2026 systematic review of web scraping published in Data & Knowledge Engineering examines scraping methods, performance metrics, emerging language-model-assisted techniques, and legal and ethical issues. The review emphasizes that web scraping transforms online information into structured data, while also requiring attention to privacy and responsible collection practices [10]. For ScholarNexus, this supports treating extraction quality and responsible access as design requirements rather than optional additions.

Scraping is challenging because websites use different layouts and terminology. A deadline might be displayed in a table, a paragraph, a downloadable notice, or a separate application page. Some sites render content dynamically, while others change their HTML structure without warning. An extractor designed for one layout may fail when the source changes. Automated extraction can

also confuse an announcement date with an application deadline or interpret an award amount incorrectly.

A reliable workflow should validate extracted values before they are displayed as confirmed facts. Useful controls include field-level checks, date parsing, duplicate detection, source URLs, last-checked timestamps, and a review queue for incomplete records. If an extraction process cannot identify a deadline or income limit with sufficient confidence, the field should remain unknown and the student should be directed to the source. The system should not invent missing values.

2.8 Data Normalization, Quality, and Freshness

Data normalization converts records from different sources into a consistent schema. For example, scholarship providers may describe academic performance using percentage, grade point average, or a pass requirement. Income thresholds may use different formats, and deadlines may appear in different date styles. A normalized database should preserve the original wording where needed while also storing validated, comparable values when conversion is safe.

Important data-quality dimensions include accuracy, completeness, consistency, uniqueness, timeliness, and traceability. Accuracy means that a stored value reflects the source. Completeness concerns whether important fields are present. Consistency means that equivalent fields follow a common representation. Uniqueness helps avoid duplicate listings. Timeliness indicates whether the information is current. Traceability enables a user or administrator to locate the original source.

Freshness is especially important for scholarship data because eligibility rules and application periods can change. A record that was correct last year may be misleading during the current academic year. A periodic refresh process can revisit known sources, compare new content with stored values, and flag records that may have changed. A visible 'last checked' date can help users judge whether further verification is needed.

The database should also preserve the distinction between confirmed values and inferred or unverified values. If the source does not mention a family income limit, the system should not display a guessed rupee amount. If two sources conflict, the system should prioritize the authoritative provider and flag the conflict. Such controls help prevent misleading recommendations and support user trust.

2.9 Eligibility Matching and Explainability

Eligibility matching evaluates whether a student's profile satisfies the published conditions of a scholarship. Typical profile fields may include education level, course or discipline, year of study, state or domicile, marks, family income, and any other condition explicitly required by a scheme. The system should request only the information needed for the matching task and explain why each field is collected.

A straightforward matching process separates conditions into three states: satisfied, not satisfied, and unknown. 'Satisfied' means the provided profile meets the known condition. 'Not satisfied' means a known value contradicts the condition. 'Unknown' means information is missing, ambiguous, or cannot be reliably compared. This three-state approach is safer than treating missing data as a match or a failure.

For example, a scholarship may require at least 60 percent marks and annual family income below a stated threshold. If a student's entered marks meet the threshold but family income has not been provided, the system should not claim full eligibility. It can indicate that the academic condition appears to match and that income eligibility remains to be checked. This makes the recommendation more transparent and actionable.

Explainability can be supported through a short reason summary, a checklist of criteria, and a link to the original provider. It should also distinguish 'potentially eligible' from 'officially verified'. The provider may impose additional requirements, document checks, merit ranking, quotas, or selection procedures that cannot be established from a profile alone.

2.10 User Interface and Student Support

A scholarship navigator must make information understandable to users with different levels of digital experience. A clear interface should prioritize essential fields, use readable labels, allow filtering, and avoid forcing students to open every listing merely to discover its deadline or minimum marks. Search results can show the scholarship name, provider, award information, deadline, and a brief eligibility summary when these values are known.

A well-designed interface should handle empty, incomplete, and stale data gracefully. If no exact matches are found, it can explain which profile conditions limited the results and allow the student

to broaden the search. It should not hide potentially useful listings simply because one optional field is missing. Instead, it can separate strong matches from opportunities that require additional verification.

A help or chatbot feature, if implemented, can explain how to use the platform, define common scholarship terms, and direct users to relevant pages. It should not invent scholarship rules or provide definitive legal or financial decisions. Answers about deadlines, award amounts, and eligibility should be grounded in stored records and source links. Where information is uncertain, the assistant should recommend checking the official provider.

2.11 Comparative Analysis of Existing Approaches

The reviewed systems and techniques solve related but distinct parts of the problem. Centralized portals support coordinated scheme administration; rule-based expert systems evaluate explicit conditions; content-based recommenders personalize results; crawlers discover pages; and scrapers convert page content into structured data. ScholarNexus brings these ideas together into a student-oriented discovery workflow.

Approach	Main contribution	Limitation relevant to ScholarNexus
Centralized scholarship portal	Combines participating schemes and application services in one platform.	May not include every institution-specific or private opportunity.
Rule-based expert system	Checks explicit conditions and can explain eligibility decisions.	Depends on complete, accurate, consistently represented rules.
Content-based recommender	Matches scholarship attributes to an individual profile.	Quality is limited by metadata and profile completeness.

Collaborative filtering	Uses patterns across multiple users to personalize suggestions.	Needs interaction data and careful privacy treatment; may suffer from cold start.
Web crawler	Discovers and revisits relevant pages from selected sources.	Does not by itself extract or validate scholarship fields.
Web scraper	Extracts selected details from page content into structured records.	Can break when layouts change and may misread ambiguous values.
Manual searching	Allows direct inspection of original information.	Requires repetitive effort and makes cross-source comparison difficult.

The comparison shows that no single technique fully addresses scholarship discovery. The proposed system combines complementary components while retaining source links and explicit uncertainty handling. Its value depends not only on collecting many listings but also on presenting trustworthy and useful information.

Chapter 3: SYSTEM DEVELOPMENT

1. Use Case Diagram

The Use Case Diagram illustrates the interaction between the system's actors, such as Students and Administrators, and the various functionalities provided by the Scholarship Navigator system. It represents the functional requirements of the system from the users' perspective.

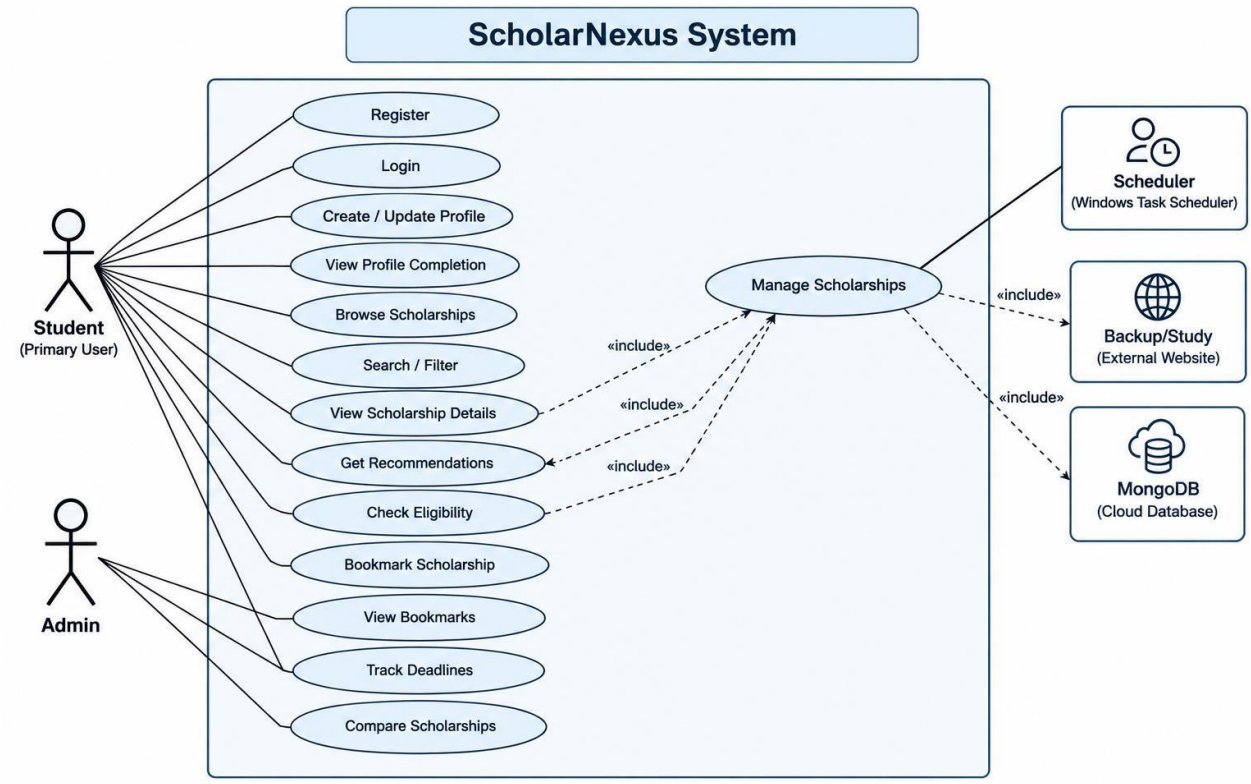


Figure 3.1 : use case diagram

2. DFD Level 0 (Context Diagram)

The Level 0 Data Flow Diagram (DFD) represents the Scholarship Navigator system as a single process and shows its interaction with external entities. It illustrates the flow of data between students, administrators, scholarship source we bsites, and the system

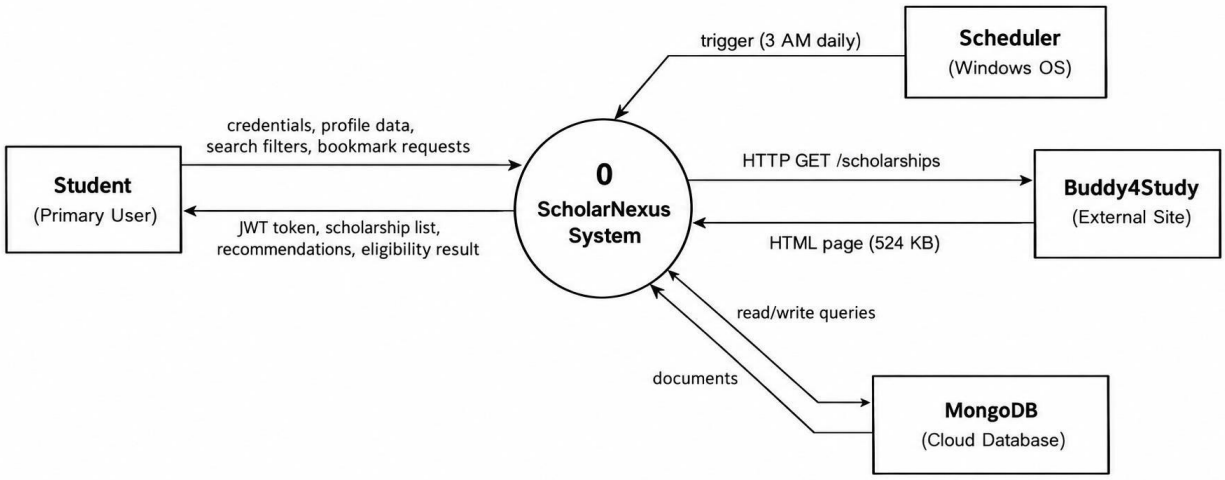


Figure 3.2 : DFD level 0

3. DFD Level 1

The Level 1 Data Flow Diagram provides a detailed representation of the major processes within the Scholarship Navigator system. It shows how scholarship data is collected, processed, stored, matched with student profiles, and presented as recommendations.

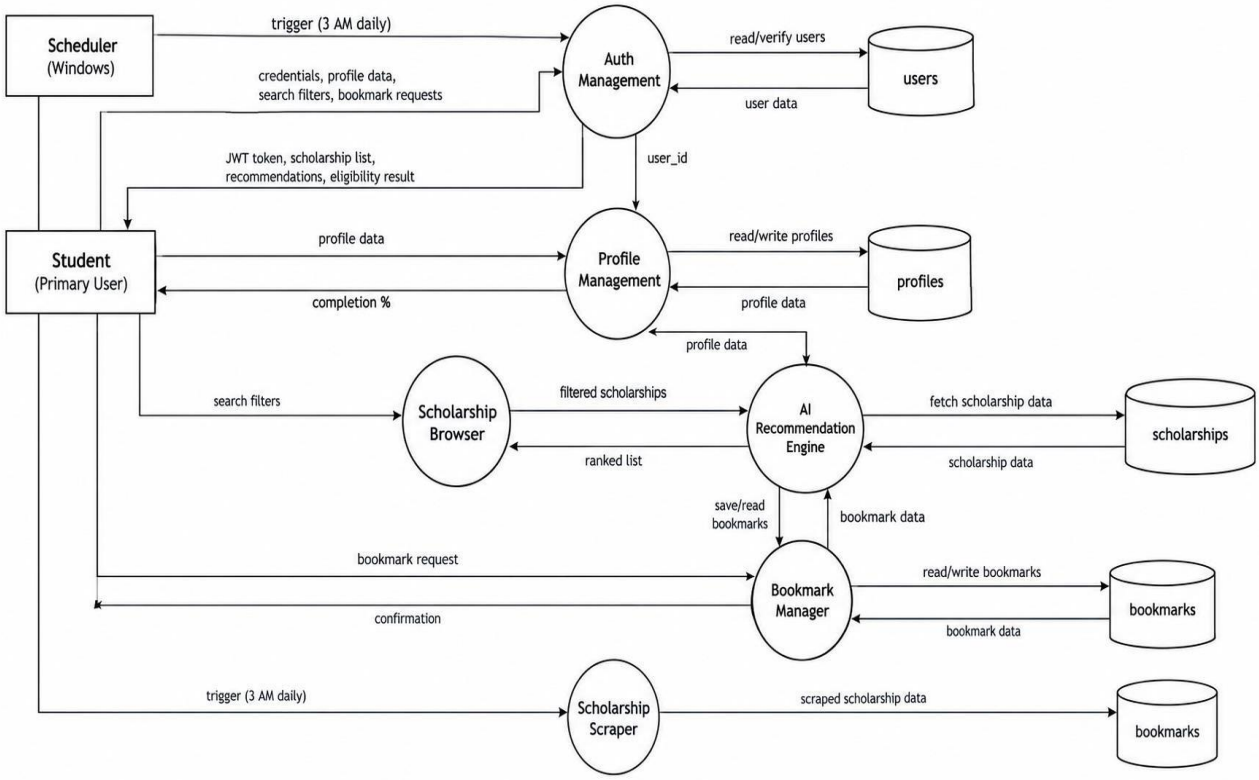


Figure 3.3 : DFD level 1

4. State Diagram

The State Diagram represents the different states of a scholarship record throughout its lifecycle, from discovery and data extraction to validation, publication, and archiving. It also illustrates the transitions between states based on system operations and administrative actions.

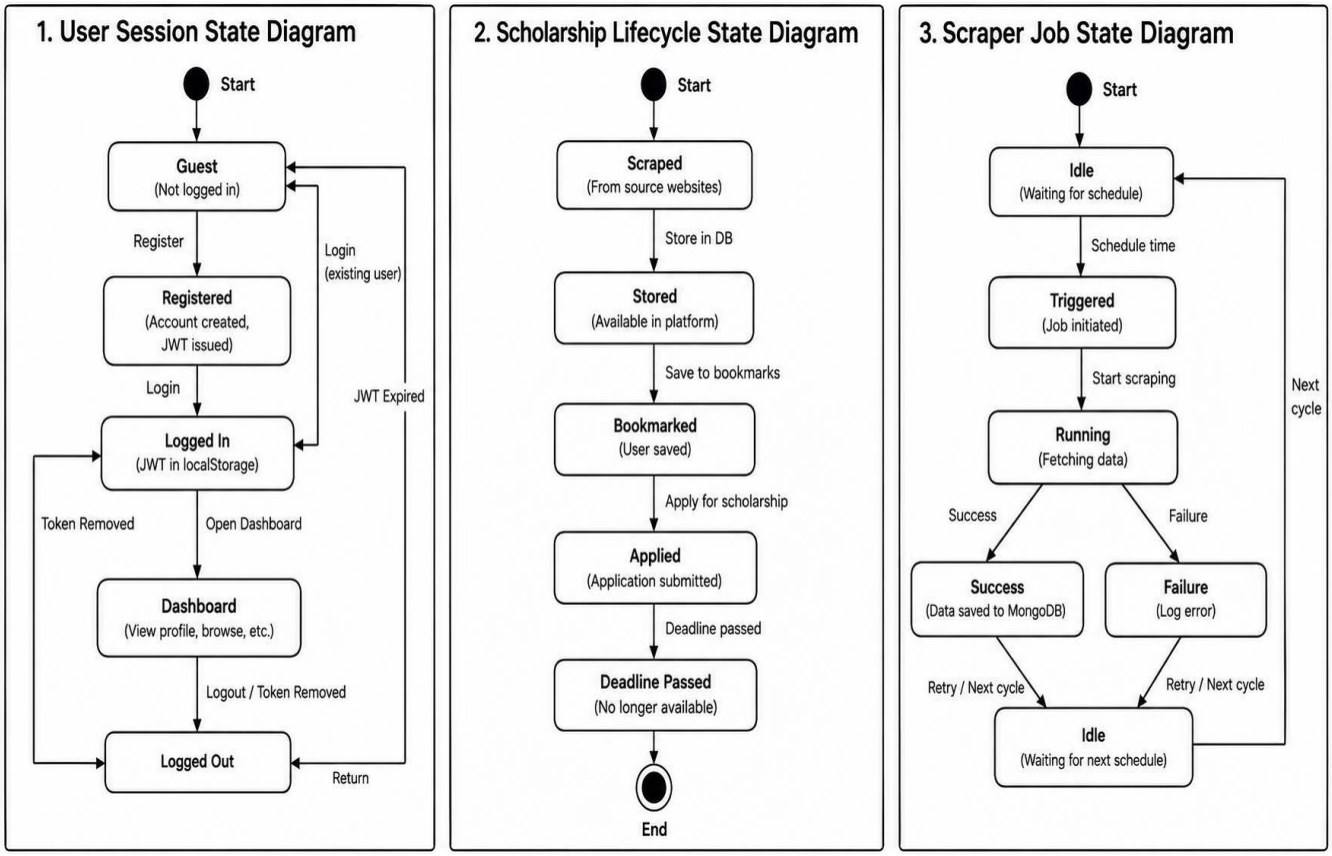


Figure 3.4: state diagram

5. Sequence Diagram

The Sequence Diagram illustrates the time-ordered interactions between the student, user interface, profile manager, matching engine, scholarship database, and recommendation ranker. It shows how these components communicate to generate and display suitable scholarship recommendations.

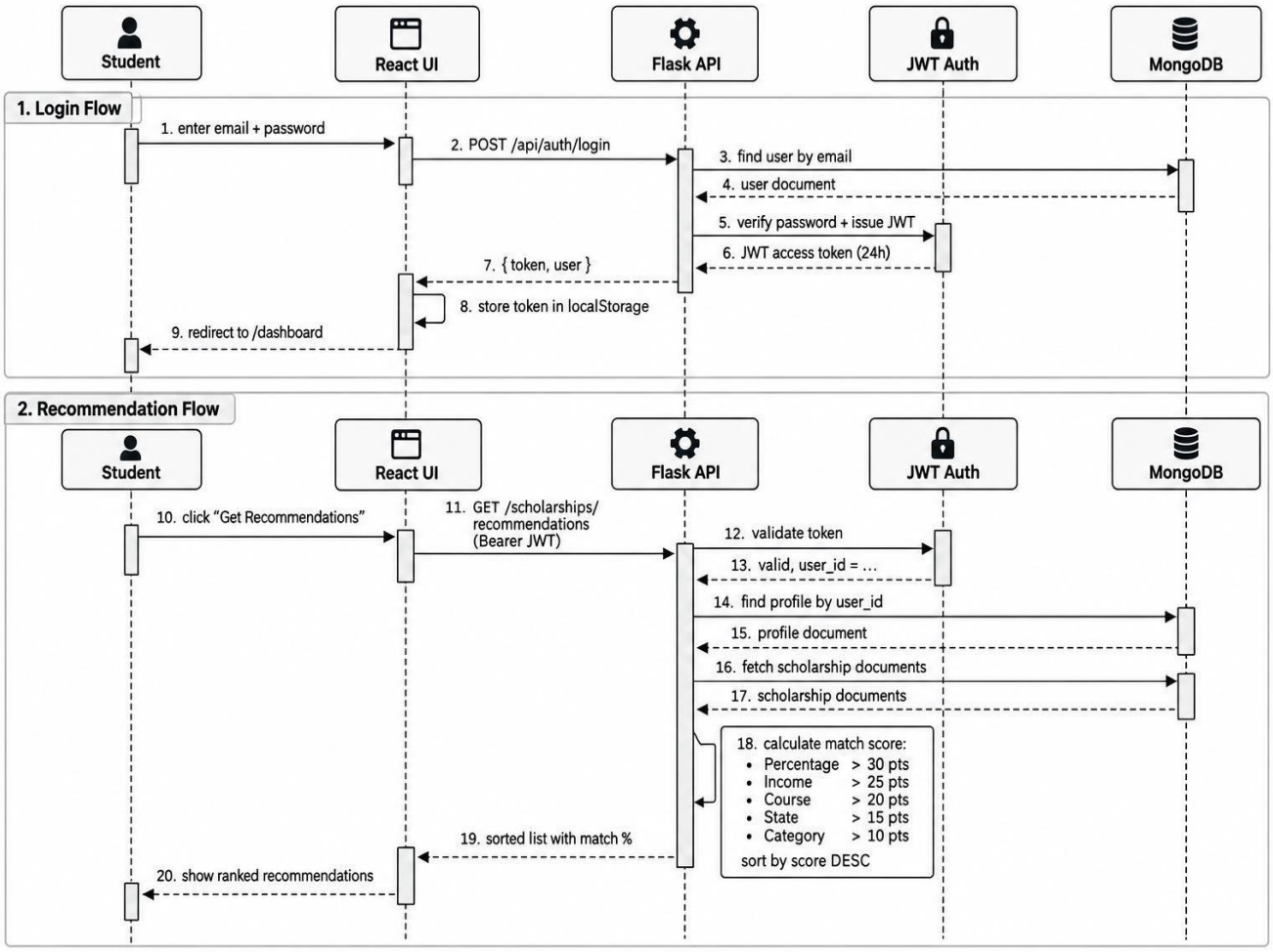


Figure 3.5: Sequence Diagram

6. Activity Diagram

The Activity Diagram represents the workflow of the Scholarship Navigator system, from collecting scholarship information and validating data to checking student eligibility and displaying recommendations. It also shows decision points that determine whether records are published, flagged for review, or filtered out.

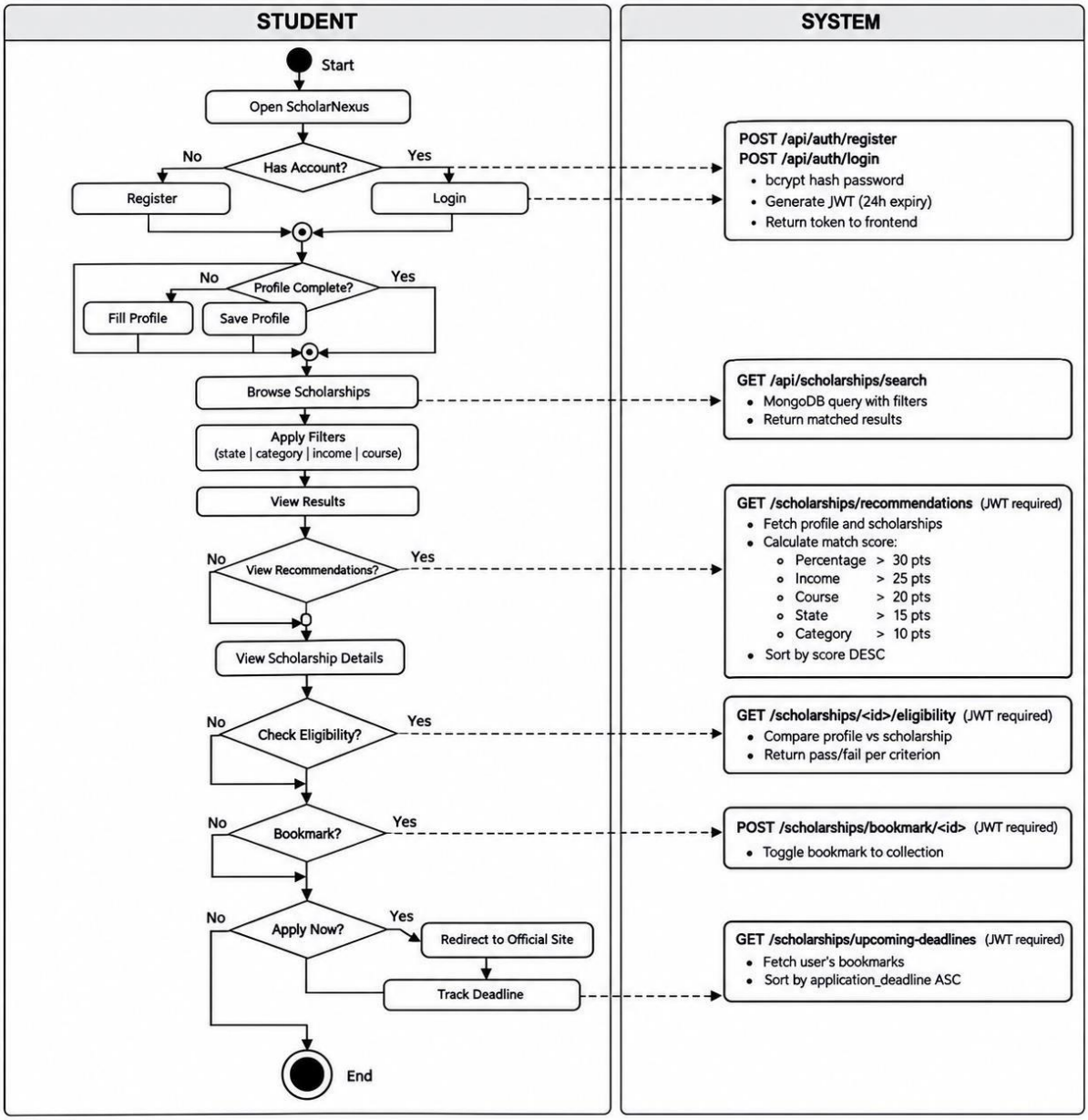


Figure 3.6: activity diagram

7. Database Design / ER Diagram

The Entity-Relationship (ER) Diagram represents the database structure of the Scholarship Navigator system, including entities such as Student, Scholarship, Source, Admin, and Match_Result. It illustrates the relationships between these entities and identifies their primary and foreign keys to support organized data storage and retrieval.

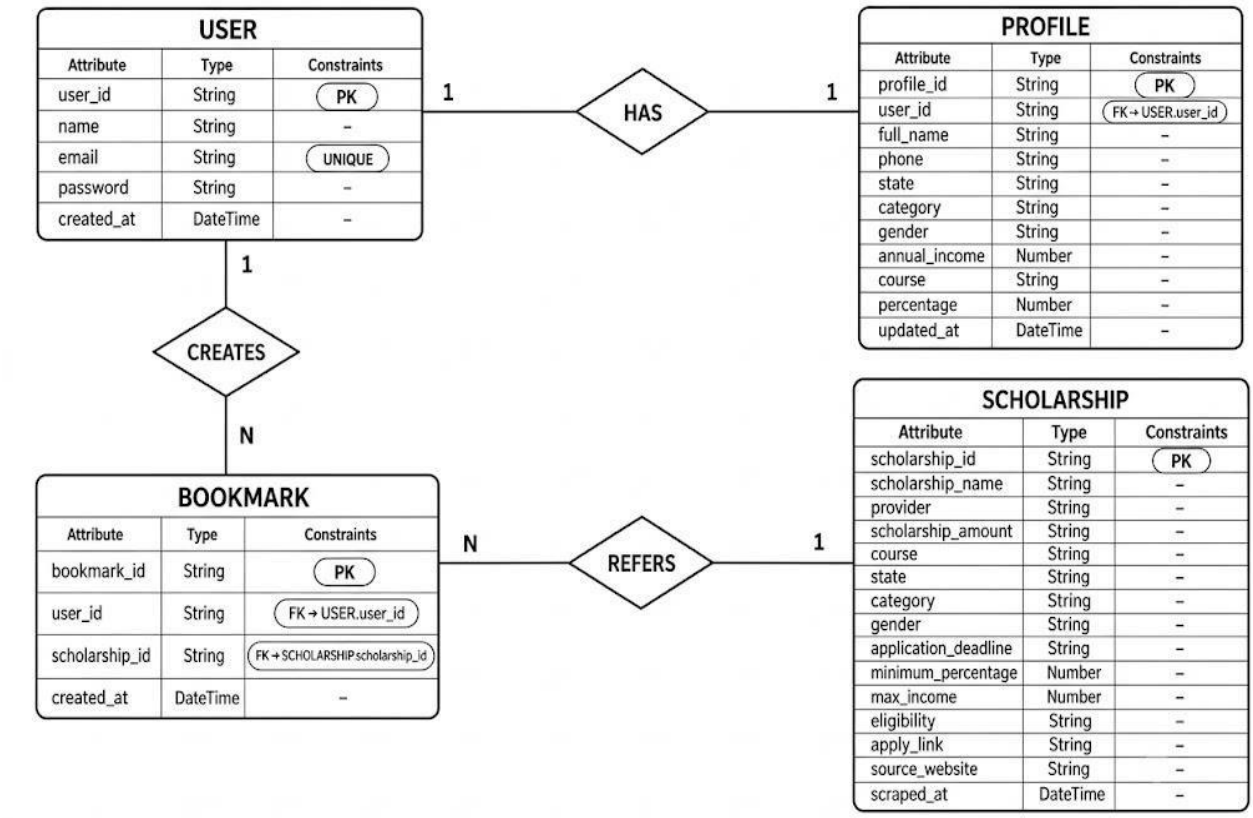


Figure 3.7 : database diagram

REFERENCES (TNR 16)(IEEE Format)

1. ***Paper Name*: Author(s), "Title of the Paper," *Journal/Conference Name*, Year, Pages.**

2. ***Book Name*: Author(s), *Book Title*, Edition, Publisher, Year.**

3. ***Book Chapter*: Author(s), "Chapter Title," in *Book Title*, Editor(s), Edition, Publisher, Year, Pages.**

4. ***Website*: Author/Organization, "Title/Topic," Website Name, URL, Accessed on: Date.**

Example with proper numbering:

1. **J. Smith and R. Brown, "Machine Learning in Healthcare," *International Journal of AI Research*, 2022, pp. 45–60.**

2. **A. Kumar, *Discrete Mathematics and Its Applications*, 8th ed., McGraw-Hill, 2019.**

3. **M. Patel, "Graph Theory Basics," in *Handbook of Discrete Mathematics*, Ed. R. Gupta, Springer, 2021, pp. 120–140.**

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Signature(s) of Students

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